INCH-POUND
MIL-C-70701B (AR)
21 December 1993
SUPERSEDING
See (6.6)

MILITARY SPECIFICATION

CABLE ASSEMBLY W8, FIRE CONTROL PROCESSOR: 9397464

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for cable assembly W8, fire control processor (cable assembly), hereinafter called "cable assembly," used on Gun, Anti-Aircraft Artillery, Self-Propelled, 20MM, M163A2, and Towed Vehicle, 20MM, M167A2. The cable assembly (located within the fire control processor) provides connections to the fire control processor (FCP) circuit boards. The cable assembly contains a 3.45 inch by 2.86 inch double-layered relay module board with one microcircuit and four limit relays (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S. Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-P-70565	Processor, Fire Control: 9360890
MIL-C-70568	Circuit **Board, Input/Output (PIvADS): 9360881**
MIL-C-70570	Circuit Board, Servo Control (PIVADS): 9360875
MIL-B-81705	Barrier Materials, Flexible Electrostatic Free, **Heat** Sealable

STANDARDS

MILITARY

MIL-STD-109 -	Quality Assurance Terms and Definitions
MIL-STD-810 -	Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies **of** federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. **4D 700 Robbins Avenue, Philadelphia, PA 19111-5094.)** "

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360874 -	Fire Control Processor Backplane Schematic
9360890 -	Fire Control Processor Assembly
9360959 -	Fire Control Processor Backplane Assembly
9397464 -	Cable Assembly W8, Fire Control Processor
12950877 -	Requirements for Soldering Electrical and Electronic Assemblies

SPECIAL PACKAGING INSTRUCTION

P9397464 - Packaging of Cable Assembly W8

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9397464, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the cable assembly shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The cable assembly W8 shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The cable assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's + 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude operating. The cable assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The cable assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

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3.3.4.2 Non-operating (storage) . The cable assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The cable assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor **shall be electrically continuous as specified on the schematic diagram of the cable** assembly.

3.5 Performance.

3.5.1 Functional description. The cable assembly shall provide connections to the FCP input/output circuit board (IOB, MIL-C-70568) and servo control circuit board (SCB, MIL-C-70570) via the FCP backplane assembly (9360959). The cable assembly shall contain a module board to provide interruption of azimuth and elevation gun motion when limit switches are activated (see Figure 1).

3.5.2 Performance characteristics.

3.5.2.1 FCP circuit board connections. The cable assembly shall provide connections to the SCB by way of the FCP backplane assembly connector J23. The cable assembly shall provide connections to the IOB by **way** of the FCP backplane assembly connector J25. Refer to Table I (Connector J8 pin designations) for cable assembly connector J8 pin designations.

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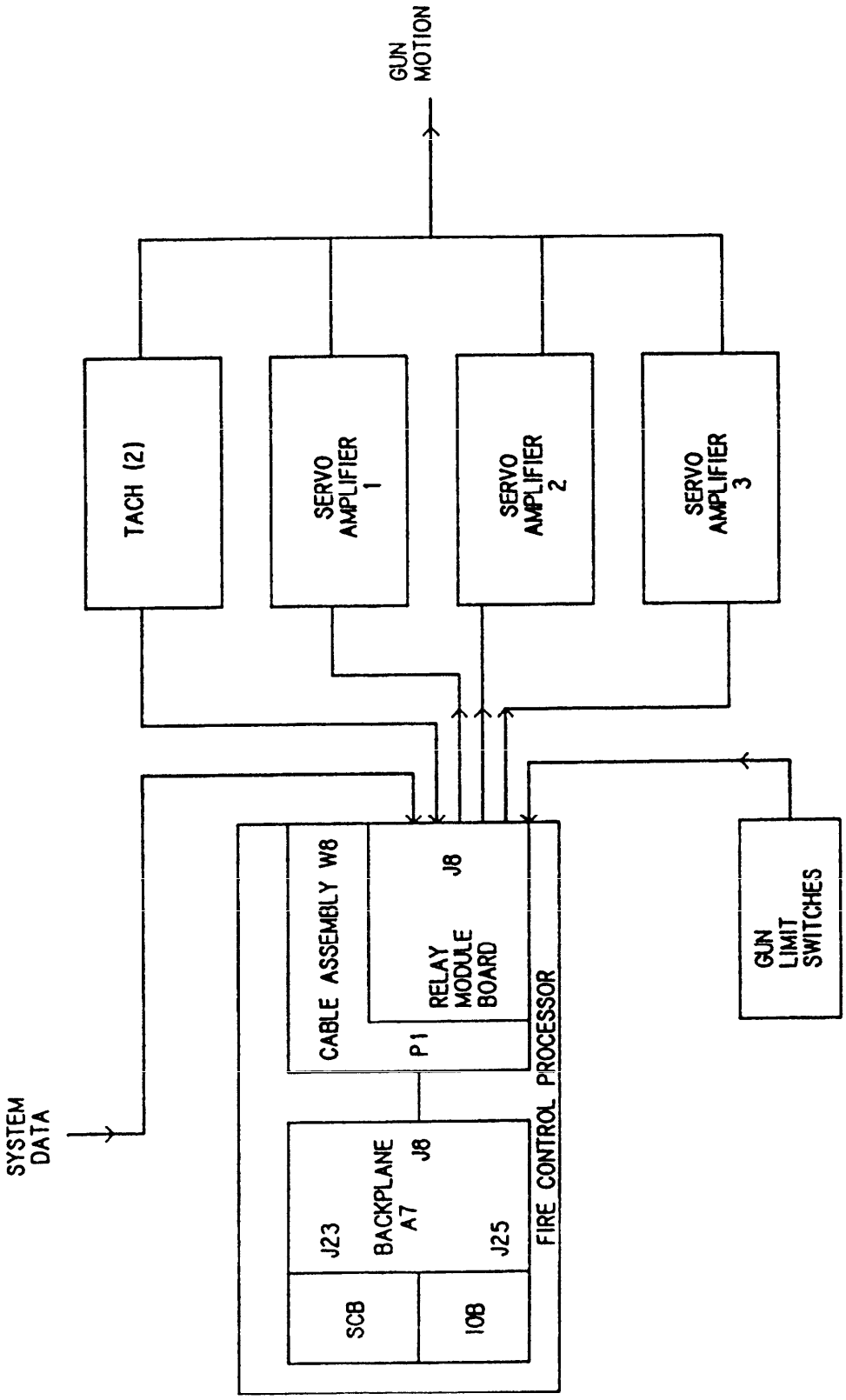


FIGURE 1. Cable assembly functional block diagram.

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TABLE I. Connector J8 pin designations.

J8 CON/PIN	CORRESPONDING P1 CON/PIN	FUNCTION	BACKPLANE/BOARD DESTINATION
J8-A	P1-27	GUN TACH AZ HI	J23-41
J8-(A)	---	CW LIMIT	RELAY MODULE (E9)
J8-B	P1-7	GUN TACH AZ LO	J23-42
J8-(B)	---	CCW LIMIT	RELAY MODULE (E8)
J8-C	---	Shield	---
J8-(C)	---	AZ LIMIT COMMON	RELAY MODULE (E7)
J8-D	P1-6	Current HI **AZ1**	J23-44
J8-(D)	P1-24	Current EL HI	J23-9
J8-E	P1-5	Current LO AZ1	J23-45
J8-(E)	P1-25	Current EL LO	J23-10
J8-F	---	Shield	---
J8-(F)	---	GUN CMMD EL HI	RELAY MODULE (El)
J8-G	P1-18	GUN CMMD **AZ1 LO**	J23-67, J25-22
J8-(G)	P1-37	GUN CMMD EL LO	J23-32, J25-57
J8-H	---	GUN CMMD AZ1 HI	RELAY MODULE (E5)
J8-(H)	---	Shield	---
J8-J	P1-4	Current AZ2 HI	J23-46
J8-K	P1-3	Current AZ2 LO	J23-47
J8-L	---	Shield	---
J8-M	P1-38	GUN CMMD AZ2 LO	J23-67, J25-22
J8-N	---	GUN CMMD AZ2 HI	RELAY MODULE (E6)

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TABLE I . Connector J8 pin designations (continued] .

J8 CON/PIN	CORRESPONDING P1 CON/PIN	FUNCTION	BACKPLANE/BOARD DESTINATION
J8-P	P1-30	DBX clock (500 KHz) HI	J25-14
J8-R	P1-lo	DBX clock (500 KHz) LO	J25-49
J8-S	P1-31	CPU to DBX Data HI	J25-8
J8-T	P1-11	CPU to DBX Data LO	J25-43
J8-U	P1-32	DBX to CPU Data HI	J25-3
J8-V	P1-12	DBX to CPU Data LO	J25-38
J8-W	---	Shield	---
J8-X	---	UP LIMIT	RELAY MODULE (E4)
J8-(X)	P1-22	GUN TACH EL HI	J23-6
J8-Y	---	DOWN LIMIT	RELAY MODULE (E3)
J8-(Y)	P1-23	GUN TACH EL LO	J23-7
J8-Z	---	EL LIMIT COMMON	RELAY MODULE (E2)
J8-(Z)	---	Shield	---
J8-AA	P1-29	AZ MOT1CW	J25-24
J8-BB	P1-9	AZ MOT1CCW	J25-59
J8-CC	P1-28	AZ MOT2CW	J25-25
J8-DD	P1-8	AZ MOT2CCW	J25-60
J8-EE	P1-21	EL MOT UP	J25-26
J8-FF	Pi-l	EL MOT DOWN	J25-61
J8-GG	P1-2	SP/$\overline{\text{TWD}}$	J23-52
J 8 - H H	P1-26	+5 Vdc RTN	J23-18

3.5.2.2 Relay module board. The relay module board shall provide servo amplifier command signal interruption when a limit switch is activated. When a limit switch activation energizes a

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limit relay, the tachometer signal of the affected axis and of the correct polarity shall be steered by diodes to the applicable servo amplifier to provide dynamic braking. The servo amplifier command signal shall then be diode-steered to allow only commands which will move the gun out of the limits. Refer to Table II (Relay module board pin designations) for connector and pin designations.

TABLE II. Relay module board pin designations.

BOARD PIN	CORRESPONDING CONNECTOR/PI N	FUNCTION	DESCRIPTION
El	J8-(F)	GUN **CMMD EL HI**	OUTPUT
E2	J8-Z	**EL LIMIT COMMON**	INPUT
E3	J8-Y	DOWN LIMIT	INPUT
E4	J8-X	UP LIMIT	INPUT
E5	J8-H	GUN CMMD **AZ1 HI**	OUTPUT
E6	J8-N	**GUN CMMD AZ2 HI**	OUTPUT
E7	J8-(C)	**AZ LIMIT** COMMON	INPUT
E8	J8-(B)	CCW LIMIT	INPUT
E9	J8-(A)	CW **LIMIT**	INPUT
E10	P1-17	**GUN CMMD EL TEST HI**	TEST POINT
E11	P1-16	**EL TACH OUT**	J23-8 INPUT
E12	P1-36	**GUN CMMD EL HI**	J23-31 INPUT
El 3	P1-13	GUN CMMD AZ TEST HI	TEST POINT
E14	P1-33	AZ TACH OUT	J23-43 INPUT
E15	P1-34	GUN CMMD AZ HI	J23-66 INPUT

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3.5.3 Test points. The cable assembly shall contain test points per Table III (Test points).

TABLE 111. Test points.

CONNECTOR-PIN	FUNCTION
J8-(F)	Test Point 1
J8-H	Test Point 3
J8-N	Test Point 2
P1-17	Test Point 4
P1-13	Test Point 5

3.6 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with requirements of Note 2 of Drawing 12950877.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that **all** products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4 .1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to **MIL-STD-109 to define quality assurance terms used herein.**

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see **4.4).**

c. Special sampling inspection (see **4.5).**

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60°F to +90°F (16°C to **+32°C)**
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg (700 to 815 mmHg)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens limited below in the quantities indicated.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which representative sample is drawn or which is inspected **100** percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection

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lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specification utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies **and subassemblies**. **All components,** assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical **Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.**

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed as specified in 4.4.2.3 and 4.5.

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4 .4.2.3 Functional tests. The tests in Table IV shall be performed on a 100% basis.

TABLE IV. Functional tests (loo percent inspection] .

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3*4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One cable assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table IV and shall then meet the tests in Table V. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE V. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. **Vibration**	**3.3.1**	**4.7.1.1**
104. **Shock**	**3.3.2**	**4.7.1.2**
105. **Altitude**	**3.3.3**	**4.7.1.3**
106. **Temperature**	**3.3.4**	**4.7.1.4**
107. **Humidity**	**3.3.5**	**4.7.1.5**
Minor:	None	**N/A**

4.5.1 **Failure of sample. Should any one item of a special** sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

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4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternative to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table VI (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE VI . Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Power supply	± 12 Vdc ± 1 percent, in accordance with Figure 2, necessary to test the relay module board and all cable assembly circuits.
2*	Digital voltmeter	Range: 0 to ± 50 Vdc. Accuracy: ± 0.1 Vdc. Input impedance: 100 Kohm minimum.
3.	**Continuity tester**	**20** Kohms input impedance.
4.	**Test** fixture	**Per Figure 2 and Table VII.**

TABLE VII. Test fixture parts list.

ITEM		DESCRIPTION
1.	Power supply	± 12 vdc ± 1%, 1 amp current limit
2.	Switch S1	Single pole, **double throw**
3.	Switch s2	Single pole, **double throw**
4.	Switch s3	**Single pole, single throw**
5.	Switch s4	**Single pole, single throw**
6.	Resistor R1	10 Kohms ± **5%, 1/4 W**
7.	Resistor R2	**10** Kohms ± 5%, 1/4 W
8.	Connector P1	P/N MS55302/113-16
9.	Connector P2	P/N M55302/27-92
10.	Connector P3	P/N M5347W22-555
11.	Connector P4	P/N MS3476W22-55PW

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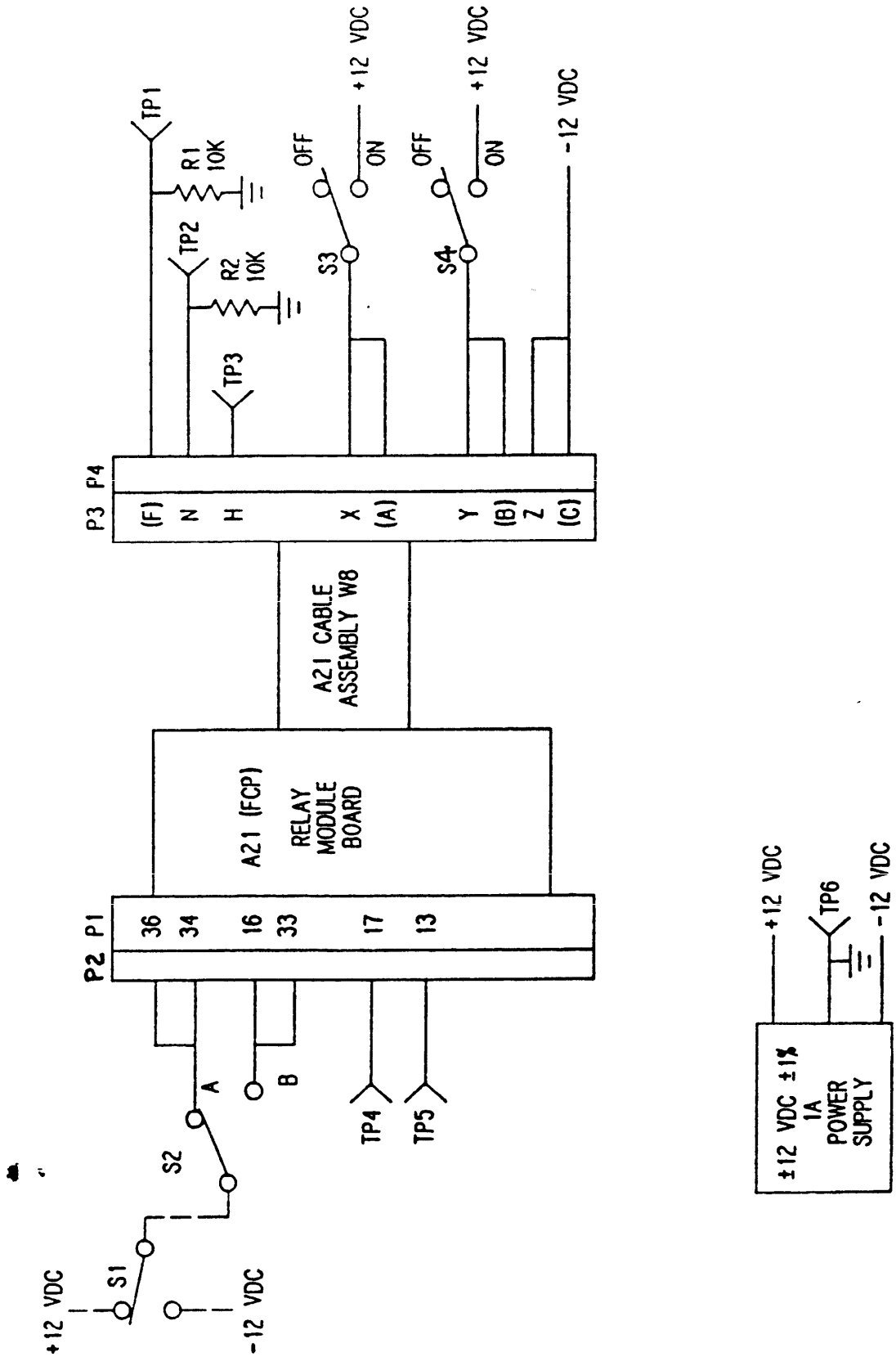


FIGURE 2. Relay module board test fixture.

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The cable assembly shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the cable assembly shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the cable assembly shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating). This test shall be performed as specified in Procedures II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.4.2 Low (operating and non-operating) This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.2.5 Humidity (non-operating) This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of-the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The cable assembly shall be continuity tested to meet the requirements of 3.5.2.1, Table I.

4.7.3.1 Relay module board. The relay module board shall be tested per Table VIII (Relay module board test procedure) using the test fixture of Figure 2 with the equipment listed in Table VI. All voltage measurements shall be made with respect to power supply common (test point 6).

TABLE VIII. Relay module board test procedure.

TEST	SWITCH POSITION S1	s2	s3	S4	VOLTAGE TP1	TP2	TP3	TP4	TP5
1	+12	A	**OFF**	OFF	+12	+12	**+12**	+12	+12
2	-12	A	OFF	OFF	-12	-12	**-12**	-12	-12
3	+12	B	OFF	OFF	0	**0**	**0**	0	0
4	-12	B	OFF	OFF	0	**0**	**0**	0	0
5	+12	A	ON	OFF	+11.4	**+11.4**	**+11.4**	+11.4	+11.4
6	-12	A	ON	OFF	0	**0**	**0**	0	0
7	**+12**	B	ON	OFF	**+1.7**	+2.5	**+2.5**	+1.7	+2.5
8	**-12**	B	ON	OFF	**0**	**0**	**0**	0	0
9	+12	A	OFF	ON	0	**0**	**0**	0	0
10	-12	A	OFF	ON	-11.4	**-11.4**	**-11.4**	-11.4	-11.4
11	+12	B	OFF	**ON**	0	**0**	**0**	0	0
12	-12	B	OFF	ON	-1.7	-2.5	**-2.5**	-1.7	-2.5

Note: All voltage measurements ± 1 vdc.

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4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.6.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9397464. The level of protection shall be as specified in the procurement documents.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cable assembly is used to interface the fire control processor with the distribution box, gun limit switches, gun tachometer, and three servo amplifiers in the PIVADS self-propelled and towed vehicles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).
Requirements for submission of first article sample.
d. Applicable stock number.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certificate of conformance for each lot or shipment of product.

6.3 Submission by contractor.

a) Inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Fern 1423 in the contract.

b) Alternative quality conformance provisions for approval Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading **U.S.** Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Key word listing.

Fire Control System
Gun, anti-aircraft artillery
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publications and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army -AR

(Project 1285-A752)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70701B (AR)	931223

3. DOCUMENT TITLE: CABLE ASSEMBLY W8, FIRE CONTROL PROCESSOR: 9397464

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6629	(2) AUTOVON DSN-880-6629
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S, Mr. V. Charles PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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